

{A relatively brief description of what I did for my independent project}

Report

The date

Prepared for

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Contents

1	Introduction	3
1.1	Some formatting tips	3
1.2	BibTeX for references	3
1.2.1	No BibTeX	4
2	Methods	4
2.1	Data	4
2.2	Modeling	4
3	Results	5
3.1	Tables	5
3.2	Figures	6
4	Discussion	7
5	References	7

1 Introduction

Make sure to edit the YAML above to include your title (embedding it in `\Large{}` makes the title page look nicer).

You might want to delete lines 24-29 above if you don't want to use **captioner** (see below). You might want to delete lines 31-32 above if you don't want to use `kable()` for any tables you might have (see below).

Also make sure to edit `report_title_page_template.tex` with the date (line 5), your name and email (line 27), and affiliation (line 31) etc. Be sure to include `report_title_page_template.tex` in the same folder where this `.Rmd` file is located.

1.1 Some formatting tips

You can use `\vspace{0.5in}` to insert a half-inch of vertical line space somewhere. You can change the 0.5 to whatever you might want. You can use `\newpage` to create a page break.

1.2 BibTeX for references

If you want to use BibTeX for managing your references in a Markdown document, you have to format your references as such and save them in a file with a “.bib” extension (most reference manager software will allow you to export a library to .bib files). Make sure to save the file in the same directory from which you knit this file. For example, the reference file for this document is called `example_refs.bib` and is listed in the YAML. You will have to change the name of the .bib file to whatever name you gave to your reference file.

To cite a reference, you use `[@ref_ID]`, which will format to (Author(s) Year). The `ref_id` should be a unique identifier in the BibTeX reference. For example, here is a BibTeX entry for the **R** software:

```
@manual{R2018,
  Address = {Vienna, Austria},
  Author = {{R Core Team}},
  Organization = {R Foundation for Statistical Computing},
  Title = {R: A Language and Environment for Statistical Computing},
  Url = {https://www.R-project.org/},
  Year = {2018},
  BdsK-Url-1 = {https://www.R-project.org/}}
```

The reference ID for this entry is R2018, so you would cite this as `[@R2018]`, which renders to (R Core Team 2018).

If you want multiple citations, you can use `[@ref_ID_1, @ref_ID_2]`. If you want to have only the year shown parenthetically without the author name(s), use `[-@ref_ID]` to get something like “were extracted with the **EGRET** package for **R** from Hirsch and De Cicco (2015)”.

1.2.1 No BibTeX

If you don't want to use BibTeX for your references, delete the following two lines from the YAML:

```
bibliography: example_refs.bib
csl: jpe.csl
```

2 Methods

2.1 Data

I'd recommend a description of your data.

2.2 Modeling

If you want your equations to be numbered, you need to enclose them with `\begin{equation}` and `\end{equation}`. If you don't want numbers you can just use `$$` at the beginning and end. For example,

```
\begin{equation}
R_t = \frac{\alpha S_t}{\mathrm{e}^{\beta S_t}},
\end{equation}
```

yields

$$R_t = \frac{\alpha S_t}{\mathrm{e}^{\beta S_t}}, \tag{1}$$

and

```
$$
R_t = \frac{\alpha S_t}{\mathrm{e}^{\beta S_t}},
$$
```

yields

$$R_t = \frac{\alpha S_t}{\mathrm{e}^{\beta S_t}},$$

If you want multiple equations in a group, you must either use the **gathered** form for centered equations, or the **aligned** form if you want them aligned (use an ampersand (`&`) in each line to indicate the point of alignment). Add a double backslash (`\\`) at the end of a line to indicate a line break.

```

\begin{equation}
\begin{gathered}
y_i \sim \text{Normal}(\mu_i, \sigma^2) \\
\mu_i = \beta_0 + \beta_1 x_{1,i}
\end{gathered}
\end{equation}

```

yields

$$\begin{aligned} y_i &\sim \text{Normal}(\mu_i, \sigma^2) \\ \mu_i &= \beta_0 + \beta_1 x_{1,i} \end{aligned} \tag{2}$$

and

```

\begin{equation}
\begin{aligned}
&y_i \sim \text{Normal}(\mu_i, \sigma^2) \\
&\mu_i = \beta_0 + \beta_1 x_{1,i}
\end{aligned}
\end{equation}

```

yields

$$\begin{aligned} y_i &\sim \text{Normal}(\mu_i, \sigma^2) \\ \mu_i &= \beta_0 + \beta_1 x_{1,i} \end{aligned} \tag{3}$$

If you want a multi-line equation with no number, embed the equation in between `$$`, such that

```

$$
\begin{aligned}
&y_i \sim \text{Normal}(\mu_i, \sigma^2) \\
&\mu_i = \beta_0 + \beta_1 x_{1,i}
\end{aligned}
$$

```

yields

$$\begin{aligned} y_i &\sim \text{Normal}(\mu_i, \sigma^2) \\ \mu_i &= \beta_0 + \beta_1 x_{1,i} \end{aligned}$$

3 Results

3.1 Tables

Below is a table with a caption above it. The table itself consists of a `data.frame`, which is formatted for nicer output with `kable()` and `kable_styling`. You specify the caption with the `caption` argument to `kable()` as seen in the example code block below

Table 1: This is a basic table caption.

Model ID	k	ΔAIC
1	2	0.0
2	2	6.4
3	1	8.5
4	3	7.0
5	3	3.8
6	3	7.1
7	2	4.2
8	3	8.6
9	3	9.5
10	1	1.4

You can find lots of different options for formatting your tables here:

https://cran.r-project.org/web/packages/kableExtra/vignettes/awesome_table_in_html.html.

```
## create some fake data
tbl_ex <- data.frame(model = seq(10),
                     k = sample(seq(3), 10, replace = TRUE),
                     deltaAIC = c(0, round(runif(9, 1, 10), 1)))

## print table
knitr::kable(tbl_ex,
              caption = "This is a basic table caption.",
              format = "latex",
              booktabs = TRUE,
              col.names = c("Model ID",
                           "$k$",
                           "$\\Delta$AIC"),
              escape = FALSE,
              row.names = FALSE,
              align = "ccr") %>%
  kable_styling(position = "center", font_size = 10)
```

3.2 Figures

You can add a figure caption below a figure using the `fig.cap` argument at the top of an R code block. For example,

```
““{r fig_ex, fig.height=3, fig.width=5, fig.align='center', echo = TRUE, fig.cap = "An example
## an ugly plot
par(mai = c(1, 1, 0, 0))
matplot(matrix(rpois(60, 10), 15, 4), las = 1, type = "l",
            xlab = expression(paste("Something here (",hat(italic(y)),")")),
            ylab = expression(paste("Another thing (",italic(e)),")"))
““
```

```
## an ugly plot
par(mai = c(1, 1, 0, 0))
matplot(matrix(rpois(60, 10), 15, 4), las = 1, type = "l",
  xlab = expression(paste("Something here (",hat(italic(y)),")")),
  ylab = expression(paste("Another thing (",italic(e)),")"))
```

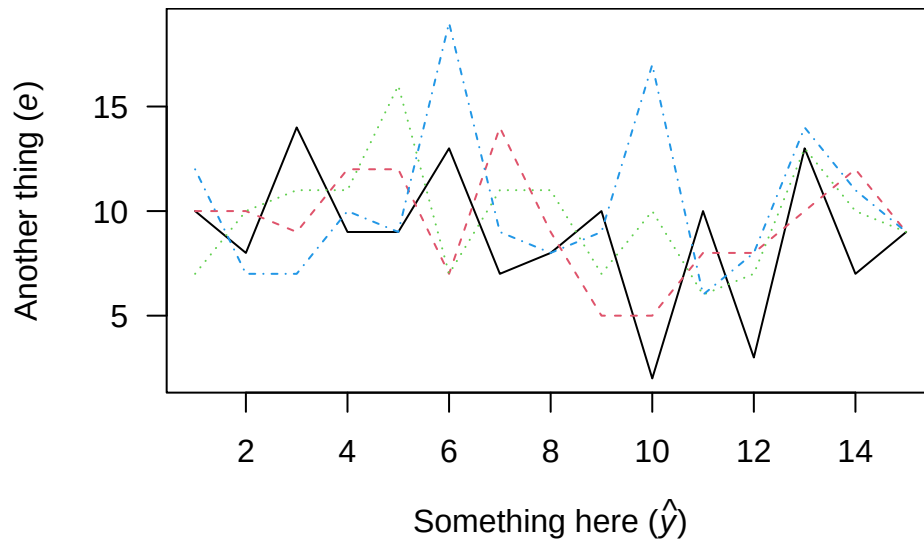


Figure 1: An example of a not very nice figure.

4 Discussion

Here is another citation (Buhle et al. 2009) to help fill out the Reference section below.

And some more text to make another paragraph.

5 References

You can replace this text with some references or just delete it if you're using the BibTeX option, but you should leave the four lines above if you want hanging indents and some space between refs.

Buhle ER, Holsman KK, Scheuerell MD, Albaugh A (2009) Using an unplanned experiment to evaluate the effects of hatcheries and environmental variation on threatened populations of wild salmon. *Biological Conservation* 142:2449–2455. <https://doi.org/10.1016/j.biocon.2009.05.013>

Hirsch RM, De Cicco LA (2015) User guide to exploration and graphics for RivEr trends (EGRET) and dataRetrieval: R packages for hydrologi data. In: *Techniques and methods*. U.S. Geological Survey; U.S. Geological Survey, Reston, VA

R Core Team (2018) R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria